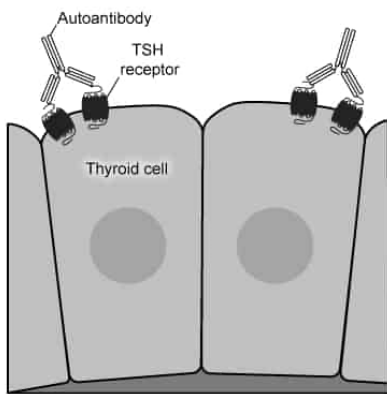


Graves' disease is a hormonal disorder in which the body produces autoantibodies specific to the thyroid gland and is characterized by heightened metabolism, as evidenced by increased sweating and weight loss. Below is a diagram of a thyroid cell in an individual affected by Graves' disease. Which statement best explains the effects of Graves' disease? (Note: Affected individuals have very low thyroid-stimulating hormone [TSH] levels.)

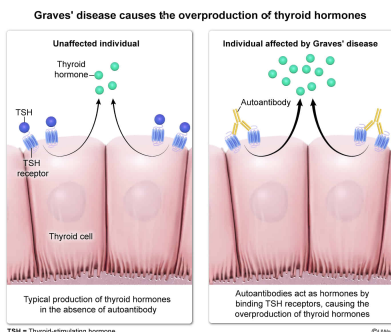


- ☐ A. The autoantibody acts as a hormone, and positive feedback control occurs on the autoantibody. [12%]
- ☒ B. The autoantibody acts as a hormone, and affected individuals experience overproduction of thyroid hormones. [60%]
- ☐ C. TSH receptors are stimulated by thyroid hormones. [2%]
- ☐ D. Affected individuals experience decreased thyroid hormone production. [24%]

Correct

60% answered correctly

Explanation:



The **hypothalamic-pituitary-thyroid axis** controls the secretion of **thyroid hormones** from the **thyroid gland** through **endocrine signaling** as follows:

1. In response to low thyroid hormone levels or a decrease in body temperature, the **hypothalamus** secretes **thyrotropin-releasing hormone (TRH)**.
2. TRH acts on the **anterior pituitary**, which causes the release of **thyroid-stimulating hormone (TSH)**.
3. TSH binds to TSH receptors on cells of the **thyroid gland**, stimulating these cells to release **thyroid hormones**.
4. Upon stimulation by TSH, thyroid gland cells secrete two types of thyroid hormones: triiodothyronine (T_3) and thyroxine (T_4).

When released, T_3 and T_4 affect nearly every cell in the body by increasing cellular metabolism. These hormones also function in a **negative feedback loop** by inhibiting the secretion of TRH and TSH.

The included figure demonstrates part of this axis in an individual affected by Graves' disease. The question states that individuals with Graves' disease show an increase in **metabolism**, which is a function of thyroid hormones. Therefore, production of thyroid hormones must be *increased* in these individuals. Because of this, the binding of autoantibodies (ie, **antibodies** that target the body's own cells, causing a reaction) to TSH **receptors** must activate these receptors in a manner similar to TSH hormones, increasing metabolism. Therefore, the **autoantibody** acts as a **hormone**, and the patient experiences **overproduction** (*not* decreased production) of **thyroid hormones** (Choice D).

(Choice A) **Positive feedback** occurs when a stimulus causes a cellular response that *further* increases the initial stimulus. Although the information in the question indicates that the autoantibody acts as a hormone in individuals affected by Graves' disease, there is *no* evidence the production of thyroid hormones (cellular response) increases the presence of autoantibodies (the stimulus). Therefore, there is no evidence of a positive feedback mechanism exerting control over the autoantibody.

(Choice C) TSH receptors are stimulated by the hormone TSH, *not* by thyroid hormones.

Educational objective:

Thyroid hormones are hormones released by the thyroid gland when it is stimulated by thyroid-stimulating hormone. Thyroid hormones affect the body by increasing cellular metabolism.

Time spent:0 sec
Subject:
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Endocrine and Nervous Systems